

Programme Specification

Title of Course: MEng Mechanical Engineering

Date Specification Produced: January 2013

Date Specification Last Revised: September 2016

This Programme Specification is designed for prospective students, current students, academic staff and potential employers. It provides a concise summary of the main features of the programme and the intended learning outcomes that a typical student might reasonably be expected to achieve and demonstrate if he/she takes full advantage of the learning opportunities that are provided. More detailed information on the teaching, learning and assessment methods, learning outcomes and content of each module can be found in Student Handbooks and Module Descriptors.

SECTION 1: GENERAL INFORMATION

Title:	MEng Mechanical Engineering
Awarding Institution:	Kingston University
Teaching Institution:	Kingston University
Location:	Roehampton Vale Campus, Kingston
Programme Accredited by:	Institution of Mechanical Engineers (IMechE)

SECTION 2: THE PROGRAMME

A. Programme Introduction

The Programme is aimed at preparing students who aim to work in Engineering and aspire to achieve their professional status of Chartered Engineer (CEng). The Programme embraces recent developments in education and industry. The Programme design is based on the guidelines provided by the Engineering Council UK Standard for Professional Engineering Competence (UK-SPEC), the Quality Assurance Agency (QAA) Subject Benchmark Statement for Engineering, and the Institution of Mechanical Engineers (IMechE) Academic Accreditation Guidelines.

MEng graduates should have the versatility and depth of understanding to enable them to deal with new and unusual challenges in their chosen field of engineering. They should be imaginative and creative so that they can become innovators. MEng graduates must be equipped to progress rapidly to a position of responsibility and provide technical, managerial, and entrepreneurial leadership in specialist or inter-disciplinary projects.

The programme has a reputation for academic excellence, intellectual rigour and industrial relevance and is designed to equip graduates with the knowledge, comprehension, intellectual ability and subject practical skills to become professional mechanical engineers or to follow careers in related professional areas. Employability is a key element of the Programme and hence the emphasis on communication, interpersonal and other skills that today's industry sees as enhancing employment prospects.

The first two years, levels four and five, of the MEng provides a broad foundation in Mechanical Engineering appropriate to the aims, objectives and learning outcomes of the programme. All MEng graduates require a foundation that covers the broad spectrum of engineering and this is provided for by studying topic which go beyond specialisation in Mechanical Engineering, hence professional issues, engineering practice and topics such as computing and electrical and electronic systems are all introduced here. In the final two years, levels six and seven, the Programme deals in much more depth with those topics which are more specific to a specialisation in Mechanical Engineering and encourages independence through the use of individual

and group project work, allowing students to showcase their abilities to deal with complex and unfamiliar data, problems and situations.

The Programme broadly follows four themes or threads, and these threads are built upon from level to level. Firstly the appreciation and application of the principles of engineering science through the core knowledge deepening subject areas of mechanical systems (statics and dynamics), thermodynamics and fluid mechanics are emphasised. Secondly, topics intended to widen the students' knowledge base include electrical and electronic systems, control engineering, mechatronics and computing. Thirdly there is a professional theme, introducing the students to the practice of a Mechanical Engineer, covering such topics as professional practice, project management, quality and business management. Lastly there is a strong design theme linking together the other threads and emphasising the holistic nature of modern day engineering. Hence, the MEng provides both breadth and depth with an aim to develop the ability to identify, define and solve engineering problems from first principles.

This MEng course emphasises the development of practical skills and experimentation through the extensive use of modern well equipped laboratories and workshops. Sustainability and 'Health and Safety' are threaded throughout the Programme's modules and students are encouraged to consider the impact that engineering has on their environment. There is a great deal of support available to students, both pastorally and academically, but in particular they are supported by a Personal Tutor Scheme (PTS) in which they are allocated a member of staff who, through one-to-one meetings, will assist and encourage students in their academic learning for the duration of their Course.

The MEng is offered as a four-year full-time degree course or a five-year sandwich course with an industrial placement taken between level 5 and level 6, or between levels 6 and 7. The industrial placement is strongly recommended and supported as the experience gained goes a long way in preparing students for employment and, along with the academic experience gained at Kingston University, ensures that our graduates are widely recognised as being well prepared for the world of work. Many of our graduates from the previous programme arrangement have gone on to very successful careers.

B. Aims of the Programme

The general aims of the course are:

- To equip graduates with the engineering, design, management, business and personal skills required to become professional Mechanical Engineers, as well as enabling them to follow careers in related professional disciplines.
- To fully meet the exemplifying academic benchmark requirements, for registration as a Chartered Engineer (CEng) and for Chartered Membership of the Institution of Mechanical Engineers (IMechE), by ensuring that the course is accredited by that body.

More specific aims of the course are:

- To produce graduates with a breadth and depth of knowledge and a thorough comprehension of the key aspects of Mechanical Engineering.
- To allow students to develop analytical and problem-solving skills and an ability to evaluate evidence and assumptions to reach sound judgements and communicate these effectively.
- To provide graduates to industry who have a creative approach to the solution of design problems and the requisite technical skills to realise these solutions.
- To equip students with the research skills required for postgraduate study and the employability skills required for work in the mechanical and related engineering industries.
- To furnish graduates with a firm grasp of sustainability and 'Health and Safety within the context of their discipline.
- To ensure that graduates have the ability and confidence to take on leadership in major engineering projects.
- To provide graduates who have the reflective skills to recognise the need to continually develop themselves in order to exercise their Professional judgement.

C. Programme Learning Outcomes

The programme provides opportunities for students to develop and demonstrate knowledge and understanding, intellectual skills and subject practical skills as outlined in the following tables. The learning outcomes are referenced to the QAA subject benchmarks for Engineering (2010) and the Framework for Higher Education Qualifications in England, Wales and Northern Ireland (2008).

Programme Learning Outcomes					
	Knowledge and Understanding		Intellectual skills		Subject Practical skills
	On completion of the course, students will be able to:		On completion of the course, students will be able to:		On completion of the course, students will be able to:
A1	Demonstrate knowledge and understanding of the core mechanical engineering subjects of statics, dynamics, materials, thermodynamics, fluid mechanics and design	B1	Apply fundamental theoretical principles that underpin engineering and specifically mechanical engineering, to existing, new and emerging technologies	C1	Use engineering workshop and laboratory equipment safely for manufacture and for experimental investigation
A2	Demonstrate knowledge of electrical and electronic systems, mathematical and computer models, control systems and basic manufacturing processes.	B2	Use mathematics as a tool for solving complex problems from first principles, communicate the results, concepts and ideas	C2	Undertake practical work and analyse the data obtained for use in planning and design
A3	Show a knowledge of broader technical and non-technical engineering subjects	B3	Think creatively and imaginatively to solve design problems	C3	Use a range of technical equipment and instruments, gaining an understanding of their underlying technology
A4	Relate management and business applications to mechanical engineering	B4	Manage projects, people, resources and time, taking account of legal and statutory requirements, risk, safety, quality and reliability.	C4	Use computer technology to assist with information retrieval, management and problem solving
A5	Demonstrate their understanding of the importance of Health and Safety in the engineering industry	B5	Demonstrate a positive attitude to learning that encourages continuing professional development throughout their careers	C5	Comply with Health and Safety regulations within the work place and as they apply to mechanical design
A6	Relate all their studies to a knowledge and understanding and promotion of sustainability and have an awareness of the environmental impact of their industry	B6	Recognise the importance of professional bodies and the professional conduct expected of Chartered Engineers		

Key Skills					
	Self-Awareness Skills		Communication Skills		Interpersonal Skills
AK1	Take responsibility for own learning and plan for and record own personal development	BK1	Express ideas clearly and unambiguously orally and in writing	CK1	Work effectively with others in a group
AK2	Recognise own academic strengths and weaknesses, reflect on performance and respond to feedback	BK2	Present, challenge and defend ideas and results effectively orally and in writing	CK2	Work flexibly and respond to change
AK3	Organise effectively, agreeing and setting realistic targets, accessing support and managing time to achieve targets	BK3	Listen actively and respond appropriately to ideas of others	CK3	Discuss and debate with others and make concessions to reach agreement
AK4	Work effectively with limited supervision	BK4	Prepare reports in prescribed and recommended forms	CK4	Give, accept and respond to constructive feedback
	Research and information Literacy Skills		Numeracy Skills		Management & Leadership Skills
DK1	Search for and select relevant sources of information	EK1	Collect data from primary and secondary sources and use appropriate methods to manipulate and analyse these data	FK1	Determine the scope of a task or project
DK2	Critically evaluate information and use it appropriately	EK2	Present and record data in appropriate formats	FK2	Identify resources needed to undertake the task or project and schedule and manage the resources
DK3	Apply the ethical and legal requirements in the access and use of information	EK3	Interpret and evaluate data to inform and justify arguments	FK3	Show the ability to successfully complete and evaluate a task or project, revising the plan where necessary
DK4	Accurately cite and reference information sources using the recommended standard method	EK4	Be aware of issues of selection, accuracy and uncertainty in the collection and analysis of data	FK4	Motivate and direct others to enable an effective contribution from all participants
DK5	Use software and ICT as appropriate				
	Creativity and Problem Solving Skills				
GK1	Apply scientific and other knowledge to analyse and evaluate information and data and to find solutions to problems				
GK2	Work with complex ideas and justify judgements made through effective use of evidence				

Teaching / Learning Methods and Strategies

The range of learning and teaching methods and strategies include staff-student contact along with a strong encouragement to develop and use appropriate Technology Enhanced Learning techniques and methods:

- Lectures
- Computer workshops/laboratories (staff or student (e.g. PAL) led)
- Group tutorials
- One-to-one tutorials
- Seminars
- Problem solving classes
- Directed reading (texts and work books: hard or e-copy)
- Directed research projects
- Visits to / by outside organizations

Assessment Strategies

The assessment strategies employed are designed to include formative and summative assessments which test the learning outcomes of the course using the following mechanisms:

- Written examinations/tests
- Multiple choice tests
- Short answer tests
- Practical laboratory sessions
- Data interpretation exercises
- Design exercises
- Group presentations
- Individual presentations
- Group Seminars
- Dissertation
- Individual reports
- Group reports
- Researched literature surveys
- Simulation exercises
- Case studies
- Research
- Computer-aided assessment

Teaching/Learning Methods and Strategies and Curriculum

- Knowledge and understanding of the discipline of Mechanical Engineering will be developed progressively from level to level. In Level 4 students will be equipped with the knowledge and skills to study more advanced topics later in the course.
- Students make progressive use of more primary, research based sources of information, and by the end of their course will have developed the skills to analyse and appraise original sources, assemble data from various sources, solve problems and carry out an individual piece of work from planning, through analysis and the design, to a solution. This proof of concept will be documented and presented in an individual capstone project at level 6 and within a major group project at level 7.
- Throughout the degree emphasis is placed on developing team work skills, written and oral communication and presentation skills, data handling and analysis skills, a range of ICT skills and independent learning skills, which are supported by the SEC Study Skills drop-in sessions.

Assessment

Assessment is regarded as an integral part of the learning and teaching strategy, with ample opportunities given to students for formative assessment with rapid feedback, particularly pronounced in the laboratory based activities and problem solving tutorial sessions, many of which feed forward into the summative assessments.

- Key skills developed during study of the course will be assessed within the various types of assessment employed (for example, the use of ICT is a normal expectation in the preparation of written work, reports etc; data handling and analysis is inherent in many of the activities as well).

The summative assessments are a mixture of coursework, consisting of a range of workshop/laboratory based assignments and in class tests, and end of module written assessments, usually in the form of unseen examinations. Each module carries a final grade, which is made up of the marks for course work and end of module assessments. The contribution of the individual assessments to the module total and the requirements to pass each module are detailed in the module descriptors.

D. Entry Requirements

The minimum entry qualifications for the programme are:

From A levels:	320 points to include A2 mathematics at grade C plus two suitable science subjects.
BTEC:	Distinction, Distinction, Merit (DDM) from an engineering-related BTEC Extended Diploma including Merit for Mathematics and Further Mathematics.
Plus:	GCSE (A*-C) minimum of 5 subjects including English Language and Mathematics.

A minimum International English Language Testing System (IELTS) score of 6.0 (min 5.5 in Speaking, Writing, Listening and Reading) or equivalent is required for those for whom English is not their first language.

E. Programme Structure

This Programme is offered in full-time and sandwich modes, this leads to the award of MEng Mechanical Engineering. Intake to both modes of study is normally in September.

Entry to the full-time and sandwich programme is normally at level 4 with A-level or equivalent qualifications (See section C). Transfer from a similar programme is possible at advanced entry levels with passes in comparable MEng preceding level modules; this is at the discretion of the Course Team and subject to the limitations and guidance of the accrediting professional body.

i. Professional and Statutory Regulatory Bodies

The Institution of Mechanical Engineers (IMechE).

ii. Work-based learning, including sandwich programmes

Work placements are actively encouraged. Students registered on the sandwich route must complete at least levels 4 and 5 before undertaking at least 36 weeks of suitable supervised work experience with an approved employer. It is the responsibility of individual students to source and secure such placements, but the Faculty and School offers considerable assistance in finding placement through the Faculty placements administrator and the School's placement tutors. During their placement students will be visited at least once by a member of the School of Mechanical and Automotive Engineering. The placement is assessed and successful completion is required for the degree to be awarded. The credits are not graded and do not contribute to the final classification. If a student is unable to find a suitable placement, they will have to transfer to the non-sandwich route.

Industrial placements allow students to reflect upon their own personal experience of working in an applied setting. This opportunity enables students to focus on aspects that can clearly relate theoretical concepts to

practice. Historically many sandwich placements have been reasonably well remunerated and students can often acquire industry relevant individual project proposals which directly relates their study to their industrial experience.

iii. Outline Programme Structure

Each level comprises four modules worth 30 credits. A student must complete 120 credits at each level. All students will be provided with the University Regulations and specific additions that are sometimes required for accreditation by outside bodies (e.g. professional or statutory bodies that confer professional accreditation). Full details of each module will be provided in module descriptors and student module guides.

Level 4 (all core)				
Compulsory modules	Module code	Credit Value	Level	Teaching Block
Engineering Applications and Practice	ME4010	30	4	1&2
Thermofluid & Mechanical Systems 1	ME4011	30	4	1&2
Analytical Methods, Computing & Electronic Systems	ME4012	30	4	1&2
Engineering Design, Materials & Manufacture 1	ME4013	30	4	1&2

Progression to level 5 requires passes in all four modules to give 120 credits at level 4. Students exiting the programme at this point, who have successfully completed 120 credits, are eligible for the award of Certificate of Higher Education.

Level 5 (all core)				
Compulsory modules	Module code	Credit Value	Level	Teaching Block
Thermofluid & Mechanical Systems 2	ME5011	30	5	1&2
Electronic Systems, Control & Computing	ME5012	30	5	1&2
Engineering Design, Materials & Manufacture 2	ME5013	30	5	1&2
Project Engineering and Management	ME5014	30	5	1&2

Progression to level 6 requires passes in all four modules to give 120 credits at level 5. Students exiting the programme at this point, who have successfully completed 120 credits, are eligible for the award of Diploma of Higher Education.

Level 6 (all core)

Compulsory modules	Module code	Credit Value	Level	Teaching Block
Thermofluid & Mechanical Systems 3	ME6011	30	6	1&2
Mechatronics, Dynamics & Control	ME6012	30	6	1&2
CAE with CFD & Mathematical Modelling	ME6013	30	6	1&2
Individual Project (BEng/MEng)	ME6014	30	6	1&2

Completion of Level 6 requires passes in all four modules to give 120 credits and qualify for Beng (Hons)

Level 7 (all core)

Compulsory modules	Module code	Credit Value	Level	Teaching Block
Mechanical Group Project (Meng)	ME7015	30	7	1&2
Engineering Research Techniques, Entrepreneurship and Quality Management	ME7711	30	7	1&2
Advanced Stress Analysis & Materials	ME7723	30	7	1&2
Computational Fluid Dynamics for Engineering Applications	ME7724	30	7	1&2

Completion of Level 7 requires passes in all four modules to give 120 credits and qualify for Meng

F. Principles of Teaching Learning and Assessment

This programme has been designed taking into account the Kingston University Curriculum Design Principles to help develop student learning from dependent to independent learning and encourage lifelong learners. A wide range of teaching and learning methods is utilised, allowing students to be fully engaged throughout the course.

Traditionally the programme has attracted a diverse cohort, with students coming from a wide range of backgrounds in terms of their education, age, gender, race, religion, sexual orientation and disability, this is seen as strong feature of the programme and results in the curriculum being designed to be as inclusive as possible. In particular the first year (level 4) is very general and allows everyone to develop their weaknesses and use their strengths to help their colleagues. Teaching, learning and assessment methods are constructed to align with the learning outcomes and syllabus content of the modules. The assessment regime of a module is designed to provide formative opportunities that allow students to improve their performance following feedback in preparation for later summative assessment. Key skills are developed throughout the programme, which are assessed both formatively and summatively. Students also have access to S³ (SEC Academic Skills Centre) for additional support on a drop-in basis giving students the opportunity to take responsibility for their own achievements and consequent learning. Generally the course will be delivered by instructional lectures whilst associated problem solving tutorials, laboratory practicals, industrial visits and design classes are used to enhance the lecture material. The course is devised to encourage and develop

students making them confident in their interpersonal and communication skills, as well as emphasising group work, data analysis and ICT skills. The contact hours associated with a module very much depends on the module type, but typically a module would comprise five hours per week of contact, which would include lecture, seminar/tutorial and design/practical sessions in various combinations.

A range of assessment methods are used enabling students to, in the initial stages of the programme, demonstrate their acquisition of the relevant knowledge, whilst in the later stages of the course the higher skills needed in industry, such as problem solving, synthesis and critical analysis are brought to the fore. Methods such as oral presentations, multiple choice questions, online assessment, written examinations, group and individual project reports, laboratory reports, peer making and informal question and answer sessions are all likely to be used depending on the learning setting. The use of a range of assessment methods is intended to ensure that no student is disadvantaged despite their educational background or disability and care is taken to avoid as far as possible bunching of assessment deadlines.

Whilst the work at level 4 concentrates on recall of fundamental concepts, the work at the higher levels applies this knowledge and understanding to engineering problem solving, often with significant levels of unfamiliarity and complexity. During their final years, level 6 and 7, students are expected to be able to synthesise and critically evaluate knowledge from various diverse sources. The level 6 individual project module is likely to involve the development of new information generated by the students themselves.

At level 7 students will be involved in a final “capstone” group design project. This will be industry relevant and will allow students to apply their advanced knowledge and abilities to deal with complex often open ended engineering problems where there is a strong likelihood that the students themselves will generate new information.

The teaching and learning strategies utilised in this course are formulated to cultivate key transferable skills considered central to academic, vocational and personal development. These skills underpin how students learn, their ability to recognise their own achievement and ability, to review and evaluate that achievement and identify future learning requirements.

Research Informed Teaching

The majority of the course team are either engineering research active or are involved in industry related professional activities, through KTPs or other direct involvement with industry. These activities played a major part in informing the course design and content, as did the direct input from industry through the activities of the Industrial Advisory Board.

Most of the teaching staff are also actively involved in the various Research Centres and/or Research Groups of the Faculty, or may be following interest areas of their own. Their activities take them into, amongst other areas, materials research both coatings and compound, into fire and explosion research both cause and prevention, into active control research and on through sustainable power generation to electric vehicle technology with particular success of the zero emissions electric motorbike. Many modules, such as Mechatronics, Dynamics and Control, extensively use case studies from the teaching teams own research to illustrate current issues and thinking within their area of interest.

Students are also able to and are encouraged to develop their own research skills which are a fundamental part of the curriculum throughout all levels of the programme. They are often encouraged, through project work, to work with research active staff on elements of live projects, and these research skills enable students to determine, distinguish and present appropriate evidentiary information in an argument, which are of great value to employers.

Staff also engage widely with the research and development of ideas in teaching and learning in Higher Education and into wider pedagogic issues which will then feed through to support learning in lectures and other forms of student engagement the programme, both formal and extra-curricular.

Technology Enhanced Learning

The use of appropriate technologies to enhance and develop learning is strongly encouraged. All modules will make use of the university's virtual learning environment StudySpace (Blackboard), not just as a repository but as an active learning tool, for example the use of video and audio recordings of lectures to additionally explain complex concepts and techniques is encouraged. Additional links to appropriate online information sources should be provided along with sign posts to appropriate resources, including MOOCs and OERs. However, the use of technology should not be restricted to the VLE, but should also embrace mobile technology and encourage students to use their mobile devices to enhance their own learning experience, this may well involve the use of social media or other application as additional aids to learning, again where appropriate given the nature of the particular module.

Types of contact

Contact with students and engagement with the programme takes many forms, some of the more widely encountered on the Meng course are:

Lectures

Lectures are formal staff-led sessions designed to introduce new topics and material or provide an overview of a topic for further student study. Lectures make use of various media, supplemented by material uploaded to StudySpace. The School's

academic staff is convinced that students learn better through active participation and hence lectures would generally overlap with tutorials in expecting students to be actively involved in sketching, designing and calculating.

Tutorials

Academic tutorials are provided where lecturers assist students in solving typical engineering problems and in discussing lecture material. In many modules the tutorials and lectures will be integrated as described above.

Design workshops

Workshops may be staff-led or student-led where students participate in group design work emphasising the need for effective oral communication. Design classes, case studies and workshops often integrate material from different academic areas and would include a practical real-world emphasis.

Practical sessions

Practical sessions are designed to enable students to acquire practical and analytical skills through the application of theory. The sessions will include data collection, analysis, presentation and reporting. Practical work will generally be carried out in small groups, requiring the production of individual or group reports depending on the nature of the activity.

Engineering workshop sessions

Throughout the first year of the course the students are given a basic introduction to common hand and machine tools, this helps reinforce design and manufacturing topics. In the following years of the programme students will have access to the engineering workshops in support of their project and design work.

Group work

Good team-working skills are an essential skill for graduates aspiring to work in any engineering industry; hence, teamwork plays an important role in the academic development of a Mechanical Engineering undergraduate. Group work projects throughout all three levels illustrate the value of team work, developing interpersonal skills and fostering cooperation and supportive peer relationships.

Individual project

A fundamental element of level 6 is the individual capstone project allowing students to integrate material from their programme in an independent study of a research topic. A student's research skills will be developed with the assistance of targeted lectures, as well as an assigned supervisor, encouraging students to work effectively independently, communicating their findings clearly and succinctly through graphical, oral and written presentation.

Assessment methods

Various assessment methods are adopted appropriate to each module to enable students to demonstrate their acquisition of knowledge and skills along with the development of their learning, as outlined in the module learning outcomes. The assessment regime for each module has been designed to provide formative opportunities that allow students to improve their performance, following feedback, in preparation for summative assessment. These formative opportunities will include, but not be restricted to, online assessment with immediate feedback, use of voting

system in class, feedback on presentations (where appropriate involving industrial partners), peer and self-assessment with tutorial support and discussion etc.. The development of skills is threaded throughout the programme and assessed both formatively and summatively. Some of the methods of assessment used in the course are:

- Report writing
- Individual and group project reports
- Individual and group designs
- Investigation of case studies
- Model building
- Short in-class tests
- Unseen and seen formal written examinations
- Individual and group practical laboratory reports
- Computer software and output analysis
- Individual and group oral presentations
- Posters

G. Support for Students and their Learning

Students are supported by:

- A Module Leader for each module
- A Course Director to help students understand their programme structure and provide academic support
- A Personal Tutor (PT) to provide academic and personal support
- A Student Support Officer (SSO) who provides additional pastoral and practical advice and support, especially to students encountering difficulties
- A dedicated Undergraduate Course Administrator
- An induction programme and study skills sessions at the start of each academic year
- An Academic Study Centre to provide support and advice to students on a daily 'drop-in' basis
- StudySpace – a versatile on-line interactive intranet and learning environment accessible both on-site and remotely
- A Staff Student Consultative Committee with student Course Representatives for each level
- A University Careers and Employability Service
- Comprehensive University support systems including the provision of advice on finance, regulations, legal matters, accommodation, international student support, disability and equality support.
- The Students' Union
- An Academic Team that seeks to maintain an open door policy in the spirit of supporting students.

Personal Tutor Scheme (PTS) in the School of Mechanical and Automotive Engineering

The following provides the aims and structure of the Personal Tutor Scheme (PTS) for the School of Mechanical and Automotive Engineering. It is intended that the PTS be embedded within the provision of the Meng programme.

Overall Aims

- To build a rapport between staff and students and contribute to personalising students' experience within the School of Mechanical and Automotive Engineering
- To support students in the development of their academic skills providing appropriate advice and guidance to students throughout their time at Kingston, while monitoring their progress, helping to identify individual needs and referring students to other University services as appropriate
- To help students to develop the ability to be self-reliant and confident self-reflective learners who use feedback to their best advantage
- To encourage students to reflect on how their learning relates to a wider context and their personal career progression

Allocation of Personal Tutors

- Personal tutors will be allocated during induction week
- Tutors will be allocated on a course basis where appropriate with student numbers being equally divided amongst the staff within the school
- Students will keep the same tutor throughout their course of study

There are specific aims and outcomes for each level, as the PTS is progressive and cumulative students will find that they are building on the skills developed in previous levels. Formative assessment will be provided in the form of regular feedback during meetings.

Level 4: Settling in and building confidence

Aims and Outcomes

- To assist students in making the transition to Higher Education and to generate a sense of belonging to the School of Mechanical and Automotive Engineering with an emphasis on widening participation issues
- To help students to develop good academic habits and to gain the confidence to operate successfully in a university context
- To prepare students to make the most of feedback throughout their course

Contact:

- Teaching block 1: three one-to-one meetings during induction week, weeks 2 and 6-7
- Teaching block 2: two one-to-one meetings during week 1 and week 6-7
- End of academic year individual 'wrap up' email

Embedded Module: ME4010 Engineering Applications and Practice

Outcome:	Assessment
To assist students in making the transition to Higher Education and to generate a sense of belonging to the School of Mechanical and Automotive Engineering	Formative (one to one meetings)
To help students' to develop good academic habits and to gain the confidence to operate successfully in a university context	Formative (one-to-one meetings, plus discussion and if necessary short exercises exploring such issues as email etiquette, report writing and general study skills)
To prepare students to make the most of feedback throughout their course	Formative (one-to-one meetings)

Level 5: Stepping it up and broadening horizons

Aims and Outcomes

- To help students comprehend and plan for the academic demands of level 5 and to support increasing independence
- To encourage students to look forward, to take up opportunities to develop wider skills and to take responsibility for their personal development
- To foster students' ability to build on and respond proactively to the feedback they have received
- To assist students in reflecting on the skills that they are developing and consider how they relate to employability

Contact:

- One-to-one meeting in week 1
- Email contact at the end of teaching block 1
- Individual 'wrap up' email at end of academic year

Embedded Module: ME5014 Project Engineering and Management

Outcome:	Assessment
To help students comprehend and plan for the academic demands of level 5 and to support increasing independence	Formative (one-to-one meetings, an explanation of how Level 5 modules contribute to degree classification, and any other differences in course structure and assessment procedures between Level 4 and Level 5)
To encourage students to look forward, to take up opportunities to develop wider skills and to take responsibility for their personal development	Formative (one-to-one meetings and preparation of a dissertation proposal or choice for level 6, plus discussions around planning for relevant placement activity)
To foster students' ability to build on, and respond proactively to the feedback they have received	Formative (one-to-one meetings and discussions within module tutorial/seminar sessions)
To assist students in reflecting on the skills that they are developing and consider how they relate to employability	Formative (one-to-one meetings and preparation of a Curriculum Vitae)

Level 6: Maximising success and moving on**Aims and Outcomes**

- To support students with the planning necessary to maximise success in their penultimate undergraduate year
- To encourage students to reflect on the employability skills they have developed and be proactive in moving towards a professional life and/or further study
- To help students to make best use of the feedback they have received so that they can build on their strengths and take steps to address any weaknesses

Contact:

- One-to-one meeting in week 1
- Email contact at the end of teaching block 1
- Individual 'wrap up' email at end of academic year

Embedded Module: ME6014 Individual Project (BEng/MEng)

Outcome:	Assessment
To support students with the planning necessary to maximise success in their final undergraduate year	Formative (one to one meetings)
To encourage students to reflect on the employability skills they have developed and be proactive in moving towards a professional life and/or further study	Formative (one to one meetings, along with the preparation and oral presentation of their Individual Project, the university Employability and Careers Service also provides activities which need to be signposted to students)
To help students to make best use of the feedback they have received so that they can build on their strengths and take steps to address any weaknesses	Formative (one to one meetings)

Level 7: Getting the most out of the Masters**Aims and Outcomes**

- To help students to make the transition to Masters level study and understand how to use feedback on the level 7 course.
- To encourage students to be proactive in making links between their course and their professional and/or academic aspirations
- To explore students' research aspirations
- To help students gain confidence in contributing to, and learning from, constructive peer review
- To encourage students to become part of a wider disciplinary and/or professional community
- To prepare students for the dynamic of supervision.

Contact:

- One-to-one meeting in week 1
- Email contact at the end of teaching block 1
- Individual 'wrap up' email at end of academic year

Embedded Module: ME7015 Mechanical Group Project (MEng)

Outcome:	Assessment
To help students to make the transition to Masters level study and understand how to use feedback on the level 7 course.	Formative (one to one planning meeting)
To encourage students to be proactive in making links between their course and their professional and/or academic aspirations	Formative (one to one meetings, along with the preparation and oral presentation of their Group Project, the university Employability and Careers Service also provides activities which need to be signposted to students)
To explore students' research aspirations	Formative (one to one meetings)
To help students gain confidence in contributing to, and learning from, constructive peer review	Formative (Group project meetings)
To encourage students to become part of a wider disciplinary and/or professional community	Formative (Group project meetings, one-to-one meetings to discuss employability and future development)
To prepare students for the dynamic of supervision	Formative (Group project meetings and on-to-one meeting)

H. Ensuring and Enhancing the Quality of the Course

The University has several methods for evaluating and improving the quality and standards of its provision. These include:

- External Examiners
- Boards of Study with student representation
- Annual review and development
- Periodic review undertaken at the subject level
- Student evaluation
- Moderation policies

Quality is also assured by the requirement for professional body (ImechE) reaccreditation, generally at a five year interval.

I. Employability Statement

The Mechanical Engineering course is designed equip students with the skills to work in many Engineering and related industries. Mechanical Engineers are found in a great many disciplines that require professional, intelligent and numerate individuals. This curriculum embeds the development of employability skills throughout the Course and is designed to equip students with the ability to relate the knowledge and skills that they have learnt to the real world contexts in which they may work in the future. Students are required to produce a CV early at level 5 and to improve this following feedback from Personal Tutors, placement advisors and

Careers and Employability Service staff, ensuring that students have a good, high quality CV ready for sending out to potential employers. Employability is greatly enhanced by many of the extracurricular activities that students become involved in and students are encouraged to reflect upon these activities and include them in their CV as appropriate. Students are actively encouraged to take part in the activities hosted by the School, such as Formula Student, Robotics Club, Design Challenge and the Engineering Society amongst others.

The school is helped by its strong links with industry, local, national and international and these links influence the development of our programmes, working with our Industrial Advisory Board gives a forum where industry can inform us of the views of employers regarding the essential employability skills that they would want to see developed in a Kingston University undergraduate. To that end the school strongly encourages and supports all students in applying for positions in industry for an Industrial Placement year. This would normally take place between levels 5 and 6 or between levels 6 and 7, the school emphasises the benefits to be obtained from an approved placement in industry in terms of future employability. An Industrial Placement comprises a period of at least 36 weeks with an approved employer. Students are required to maintain a log book of their activities and involvement and produce a final report on their placement, they are supported throughout the period by their personal tutor, who will visit them at their place of work on at least one occasion. The tutor will discuss progress with the student and employer and will recommend any improvements to the learning opportunities. Students fulfilling the requirements for an Industrial Placement will be awarded a Sandwich Degree on the completion of level 7.

As well as discipline specific employability skills, the more generic employability skills are also embedded within the programme. During the first year in modules such as Engineering Application and Practice, students are encouraged to consider their chosen discipline in the light of the activities of the Professional Bodies. Again in the level 5 Project Engineering and Management modules the various careers within engineering are considered and students are encouraged to reflect on the paths open to them and to develop the skills and attributes employers are looking for in graduates. These include independent learning, the ability to work in teams, time management skills, verbal and written communication skills. Part of the role of the Personal Tutor is to encourage students to develop this skills outside of the curriculum as well as within it through extracurricular activities such as volunteering, positions of responsibility within clubs and societies, student ambassadorship and sports activities.

This programme has been designed to fully meet the exemplifying academic benchmark requirements, for registration as a Chartered Engineer (Ceng).. Most graduates will aspire to careers in mechanically related industries and to becoming Chartered Engineers. Graduates develop careers in all branches of mechanical and related engineering industries both here in the UK and throughout the world; as contract and consulting engineers, within local authorities, utility, manufacturing and transport companies, government organisations and the defence industry. In many cases, students taking an industrial placement are able to secure employment with the placement organisation following graduation. The academic and key skills developed throughout an engineering course also allow graduates to follow careers in other professions such as ICT, finance, accountancy and teaching.

J. Approved Variants from the Undergraduate Regulations Compensation

Compensation is not permitted for the following modules:

ME6014 Individual Project (BEng/MEng)
ME7015 Mechanical Group Project (MEng)

Reassessment of Level 6 and Level 7 modules

Reassessment of ME6014 or ME7015, will normally be by repeat only with a new project brief.

K. Other sources of information that you may wish to consult

Engineering subject benchmark:

www.gaa.ac.uk/Publications/InformationAndGuidance/Pages/Subject-benchmark-statement-Engineering-.aspx

Professional bodies:

www.imeche.org/

Professional accreditation:

www.engc.org.uk/

www.imeche.org/

School Website:

<http://sec.kingston.ac.uk/about-SEC/schools/mechanical-and-automotive-engineering/>

Development of Programme Learning Outcomes in Modules

This table maps where the programme learning outcomes are assessed across the twelve Meng modules. It provides an aid to academic staff in understanding how individual modules contribute to the programme aims, a means to help students monitor their own learning and personal and professional development as the programme progresses, as well as a checklist for quality assurance purposes.

		Level 4				Level 5				Level 6				Level 7			
Module Code		ME4010	ME4011	ME4012	ME4013	ME5011	ME5012	ME5013	ME5014	ME6011	ME6012	ME6013	ME6014	ME7015	ME7711	ME7723	ME7724
Programme Learning Outcomes	Knowledge & Understanding	A1	F	SF		SF		SF		SF		F	SF	S		S	F
		A2			SF	SF		SF			SF	S	SF	S		S	SF
		A3	F		F	SF	F	SF	SF	F	F	F	SF	SF	S		
		A4	F			F		F	SF				F	SF	S	SF	
		A5	F			F			SF				SF	F	F		
		A6	F	F		SF	F		F	SF	F	F		SF	S	F	
	Intellectual Skills	B1		SF	SF	SF	SF	SF		SF	SF	F	F	SF		S	S
		B2		SF	SF	F	SF	SF	F	SF	SF	SF	F	S	F	S	S
		B3				SF		F	SF	F		F	SF	S	F	F	F
		B4	SF			SF		F	SF				F	S	SF		
		B5	SF			F		F	SF	F	F	F	SF	SF	F	F	
		B6	SF						SF				F	F	F		
	Practical Skills	C1	SF	F	F	SF	F	F	F		F	SF		F	F		
		C2	SF	SF	SF	SF	F		SF		SF	SF	S	F	SF		S
		C3	SF	F	SF		F		F		F	F	F	F		F	
		C4		F	SF	SF	F	SF	SF	SF	SF	S	F	S		S	S
		C5	SF	SF	F	SF	F		F	SF	F	F		F	F		

S indicates where a summative assessment occurs.

F where formative assessment/feedback occurs.

			Level 4				Level 5				Level 6				Level 7				
Module Code			ME4010	ME4011	ME4012	ME4013	ME5011	ME5012	ME5013	ME5014	ME6010	ME6011	ME6013	ME6014	ME7015	ME7711	ME7723	ME7724	
Programme Learning Outcomes (Key Skills)	Self-Awareness	AK1	F			F	F	F	F	SF	F	F	F	S	S	SF			
		AK2	SF	F	F	F	F	F	SF	SF	SF	F	F	S	S	F	F		
		AK3	F			SF			F	F	F			SF	S	F	F		
		AK4				F	F	F	F		F	F	F	S	S	SF	S	S	
	Communication	BK1	F	F		SF	SF	F	SF	F	S	S	S	SF	SF	S	F	SF	
		BK2	SF		F		F		SF	F	SF	S	S	SF	S	S			
		BK3	F			SF			F	F	F	F	F	S	S				
		BK4	SF	SF	F	SF	SF	F	SF	SF	S	S	S	SF	S	S	F	F	
	Interpersonal	CK1	SF	F	F	SF	SF	SF	SF	SF		F	SF		S				
		CK2				F	F	F		F	F		F	S	S	F	F	F	
		CK3	F		F	F	F	F		F		F	F		S	F	F	F	
		CK4	SF	F	F	F	F	F	F	SF	S	SF	SF	S	S				
	Research and Information Literacy	DK1	SF			SF	F		SF	SF	SF	F	SF	SF	SF	SF	F	F	F
		DK2	SF	F	F	F	F	F	SF	F	F	S	S	SF	SF	SF	S	S	
		DK3	F			F	F	F		F	F	F	F	SF	S	F			
		DK4	SF	F	F	SF	F	F	SF	F	S	S	S	SF	S	F			
		DK5	SF	F	SF	SF	F	SF	SF	F		S	S	S	F	F	S	S	
	Numeracy	EK1	F	SF	SF	F	SF	SF	F	F		SF	SF	SF	S	F	S	S	
		EK2	F	SF	SF		SF	SF	SF	F	F	S	S	SF	S	S	S	S	
		EK3	F	F	F	F	SF	SF	F	F	F	S	S	SF	S	S	S	S	
		EK4	SF	SF	SF	F	SF	SF	SF			S	S	SF	F		S	S	
	Management and Leadership	FK1	F		F	SF		F	F	F	F		F	S	S	F			
		FK2	F			SF		SF	F	F				SF	S	S			
		FK3	F			SF	F	F	F	F				S	S	F			
		FK4	SF	F		SF	F	F		F			SF		S				
	Creativity and Problem Solving	GK1	F	SF	SF	SF	SF	SF	SF	F	SF	S	SF	S	S		S	S	
		GK2		F			F	F	SF		SF	S	S	S	S	S	S	S	

Mapping of Learning Outcomes for Professional Body Accreditation:

EC UK-SPEC

Summary of Specific Learning Outcomes for an Integrated Meng degree.

Specific Learning Outcomes in Engineering

Graduates from accredited programmes must achieve the following five learning outcomes, defined by broad areas of learning. As set out here, the outcomes apply to accredited programmes at Integrated Meng degree level leading to Ceng registration.

The weighting given to these different broad areas of learning will vary according to the nature and aims of each programme.

Underpinning science and mathematics, and associated engineering disciplines, as defined by the relevant engineering institution

- A comprehensive understanding of the scientific principles of mechanical and related engineering disciplines.
- A comprehensive knowledge and understanding of mathematical and computer models relevant to the mechanical and related engineering disciplines, and an appreciation of their limitations.
- An understanding of concepts from a range of areas including some outside engineering, and the ability to apply them effectively in engineering projects.
- An awareness of developing technologies related to mechanical engineering.

Engineering Analysis

- Ability to use fundamental knowledge to investigate new and emerging technologies.
- Ability to extract data pertinent to an unfamiliar problem, and apply its solution using computer based engineering tools when appropriate
- Ability to apply mathematical and computer-based models for solving problems in engineering, and the ability to assess the limitations of particular cases
- Understanding of and ability to apply a systems approach to engineering problems.

Design

Design is the creation and development of an economically viable product, process or system to meet a defined need. It involves significant technical and intellectual challenges and can be used to integrate all engineering understanding, knowledge and skills to the solution of real problems. Graduates will therefore need the knowledge, understanding and skills to:

- Wide knowledge and comprehensive understanding of design processes and methodologies and the ability to apply and adapt them in unfamiliar situations.
- Understand customer and user needs and the importance of considerations such as aesthetics;
- Identify and manage cost drivers;
- Ability to generate an innovative design for products, systems, components or processes to fulfil new needs.
- Ensure fitness for purpose for all aspects of the problem including production, operation, maintenance and disposal;
- Manage the design process and evaluate outcomes.

Economic, social and environmental context

- The ability to make general evaluations of commercial risks through some understanding of the basis of such risks.
- Extensive knowledge and understanding of management and business practices, and their limitations, and how these may be applied appropriately *to strategic and tactical issues*.
- Understanding of the requirement for engineering activities to promote sustainable development;
- Awareness of the framework of relevant legal requirements governing engineering activities, including personnel, health, safety, and risk (including environmental risk) issues;
- Understanding of the need for a high level of professional and ethical conduct in engineering.

Engineering Practice

Practical application of engineering skills, combining theory and experience, and use of other relevant knowledge and skills. This can include:

- A thorough understanding of current practice and its limitations and some appreciation of likely new developments;
- Extensive knowledge and understanding of a wide range of engineering materials and components;
- Understanding of contexts in which engineering knowledge can be applied (e.g. operations and management, technology development, etc.);
- Understanding use of technical literature and other information sources;
- Awareness of nature of intellectual property and contractual issues;
- Understanding of appropriate codes of practice and industry standards;
- Awareness of quality issues;
- Ability to apply engineering techniques taking account of a range of commercial and industrial constraints.

Engineering Council Specific Learning Outcomes - Meng Mechanical Engineering

Contribution of Module Learning Outcomes (numbered) to the specific Learning Outcomes listed in UK-SPEC

Module Code	Module Title	Underpinning Science and Mathematics	Engineering Analysis	Design	Economic, Social and Environmental Context	Engineering Practice
ME4010	Engineering Applications and Practice		5	4,5,6	4,6	1,2,3
ME4011	Thermofluid & Mechanical Systems 1	1,2,3,4,5,6	1,2,3,4,5,6		1	2,4
ME4012	Analytical Methods, Computing & Electronic Systems	1,2,3,4,5,6	3,4,5,6			2,3
ME4013	Engineering Design, Materials & Manufacture 1	4,5,6	2,4,6	1,2,3	3	1,2,5
ME5011	Thermofluid & Mechanical Systems 2	1,2,3,4,5	6		1,2	3
ME5012	Electronic Systems, Control & Computing	1,3,4	2,3,4,5,6			
ME5013	Engineering Design, Materials & Manufacture 2	2	3,4,5	1,4,5,6	1,2,6	3
ME5014	Project Engineering and Management		2		1,2,4	1,2,3,4,5,6
ME6011	Thermofluid & Mechanical Systems 3	1,2,4,6	1,3,5,6	6		5
ME6012	Mechatronics, Dynamics & Control	1,3	2,3,4,5			
ME6013	CAE with CFD & Mathematical Modelling					
ME6014	Individual Project (BEng/MEng)		1,2,3,4,5	1,2,3,4,5	2,3,5	1,5
ME7015	Mechanical Group Project (Meng)	1	1,4	1,2,3,4,	1,2	1
ME7711	Engineering Research Techniques, Entrepreneurship and Quality Management	1,2	3,4		3,4,5,6	3,4,6
ME7723	Advanced Stress Analysis & Materials	1,2,4,5	1,2,3			5,6
ME7724	Computational Fluid Dynamics for Engineering Applications	1,2,3	2,3,4			4

Technical Annex

Final Award(s):	MEng Mechanical Engineering
Intermediate Award(s):	Cert HE in Mechanical Engineering Dip HE in Mechanical Engineering BEng (Hons) Mechanical Engineering
Minimum period of registration:	FT – 4 years
Maximum period of registration:	FT – 8 years
FHEQ Level for the Final Award:	Level 7
QAA Subject Benchmark:	Engineering
Modes of Delivery:	Full-time, Sandwich and Part-time
Language of Delivery:	English
Faculty:	SEC
School:	Mechanical and Automotive Engineering
JACS code:	H300
UCAS Code:	H303 (full-time, part-time) H304 (sandwich)
Course Code:	NMEKUDE4F (full-time) NMEKUDE5S (sandwich)
Route Code:	NUMEE (full-time) NYMEE (sandwich)

MEng in Mechanical Engineering

LEVEL 4

Engineering Application and Practice
ME4010
100% ica

Thermofluid & Mechanical Systems 1
ME4011
40% ica 60% ex

Analytical Methods, Computing,
Electrical & Electronic Systems
ME4012
100% ica

Engineering Design, Materials &
Manufacture 1
ME4013
70% ica 30% ex

LEVEL 5

Thermofluid & Mechanical Systems 2
ME5011
50% ica 50% ex

Electronic Systems, Control &
Computing
ME5012
50% prac ex 50% ex

Engineering Design, Materials &
Manufacture 2
ME5013
100% ica

Project Engineering & Management
ME5014
40% ica 60% ex

**Possible Industrial
Placement**

Key:
ica = in-course assessment
ex = examination
prac ex = practical exam

MEng in Mechanical Engineering

LEVEL 6

Thermofluid & Mechanical Systems 3
ME6011
50% ica 50% ex

Mechatronics, Dynamics & Control
ME6012
30% ica 70% ex

CAE with CFD and
Mathematical Modelling
ME6013
50% ica 50% ex

Individual Project (BEng/MEng)
ME6014
90% ica 10% prac ex

Alternative Industrial Placement

LEVEL 7

Mechanical Group Project (MEng)
ME7015
80% ica 20% prac ex

Engineering Research Techniques,
Entrepreneurship and Quality
Management
ME7711
100% ica

Advanced Stress Analysis & Materials
ME7723
50% ica 50% ex

Computational Fluid Dynamics for
Engineering Applications
ME7724
100% ica

Key:
ica = in-course assessment
ex = examination
prac ex = practical exam